MicroBlaze-V-Zephyr-Kria-OpenAMP

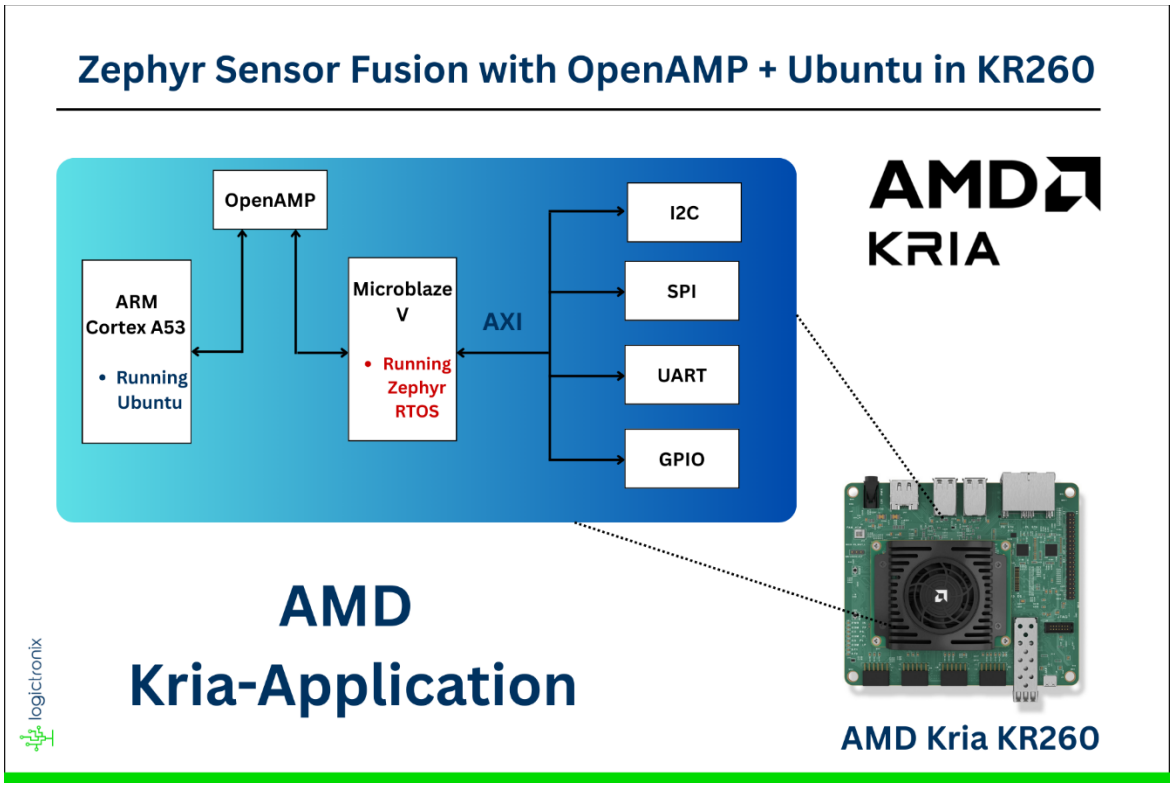
Getting Started Guide, document revision v1.0

A. Overview

This Kria Application is on Running Zephyr RTOS application for Sensor Fusion on Microblaze V along with OpenAMP on top of Ubuntu/Linux on the KR260 Board's MPSoC PS (A53) core.

This Kria application showcase the sensor fusion and multi-OS integration with the Kria Robotics Starter Kit (KR260). We are utilizing Digilent Pmod Sensors for Temperature and Accelerometer along with Pmod based UART and GPIO.

Architecture of the Kria App:



B. Preparing the KR260 and Sensor Setup

1. **Necessary Hardware - KR260 and Pmod Accessories**
   a. KR260 Board , power adapter , 16+ GB of SD Card – Purchase Link from AMD
   b. Pmod Acl – Digilent Link
   c. Pmod TMP2 – Digilent Link
   d. Pmod GPIO 8 bit LED – Digilent Link
   e. Pmod UART – Digilent Link
   f. Few Male to female jumpers – Amazon or can be purchased from any electronics shop.

2. **Preparing the KR260 Board with Ubuntu 22.04 LTS image**
   a. Follow the "**Booting Kria Starter Kit Linux on KR260**" steps from here - **ubuntu_22_04 [booting linux on KR260]**.
   b. Where you will:
      - Download the image
      - Burn SD card using Balena etcher
      - Update the boot firmware at KR260
      - And boot the KR260 with Ubuntu 22.04LTS.

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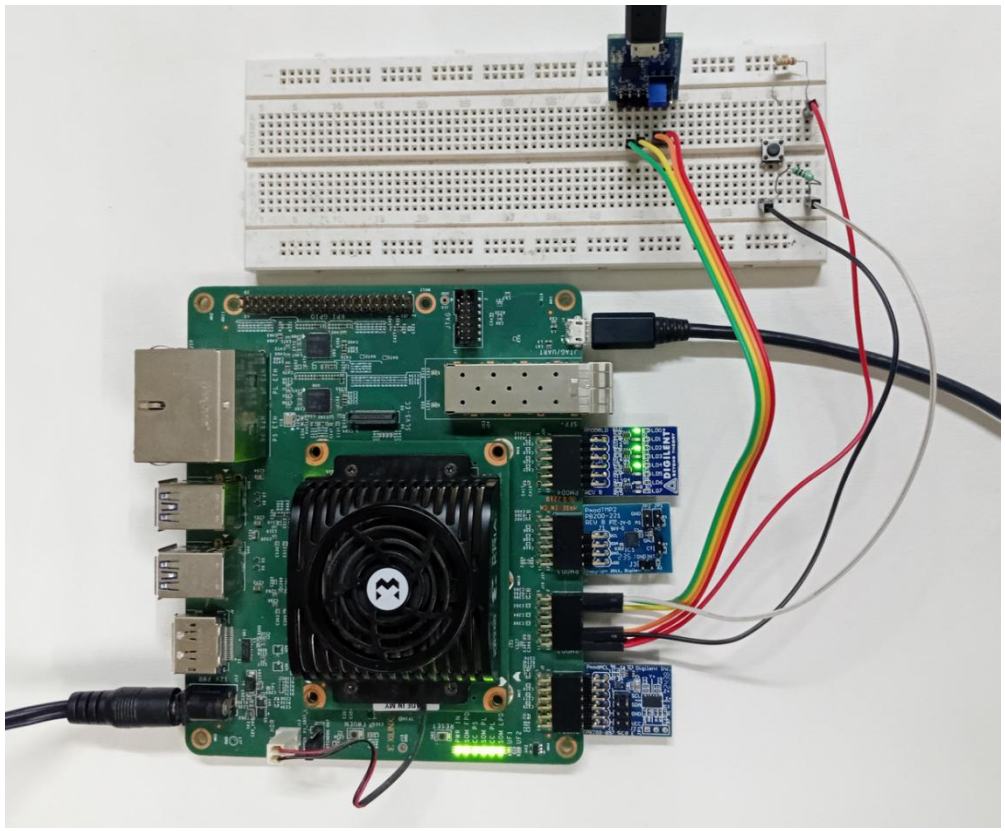
3. **Connecting the Pmod Sensors with KR260 Robotics Starter Kit:**
   Below are the details for connection of Pmod sensors and modules at KR260.
   a. Pmod Acl - Accelerometer
      - Connect to Pmod1.
   b. Pmod TMP2 – Temperature sensor
      - Connect to Pmod3.
   c. Pmod GPIO – 8 bit LED
      - Connect to Pmod4.
   d. Pmod UART and PL Reset
      - Connect to Pmod2.
      - You can follow following VIVADO Constraint section for connection.

```
# UART_txd PMOD #2 pin 2 (connect rx of the uart device to pin 2)
# UART_rxd PMOD #2 pin 4 (connect tx of the uart device to pin 4)
# Reset button → PMOD #2 pin 1
```

4. **Connect the Pmod-UART and KR260 UART on the board with PC.**
    a. You can use Mobaxterm/Teraterm in ***Windows*** or Gtkterm in ***Linux***.
    b. Connect the baudrate of 115200 for both of the UARTs

5. **KR260 Pmod sensors and connection:**



6. **Now boot the KR260 board.**
    a. After booting , you will start getting boot log from Ubuntu 22.04LTS into your PC/computer.
    b. After boot completes, you can enter the username – ubuntu and password – which you have setup in step2.

C. Clone Github Repository

Clone this github repository at the KR260-Ubuntu image by running git clone command:

```
git clone https://github.com/LogicTronixInc/MicroBlaze-V-Zephyr-Kria-OpenAMP.git
```

D. Download and install the packages

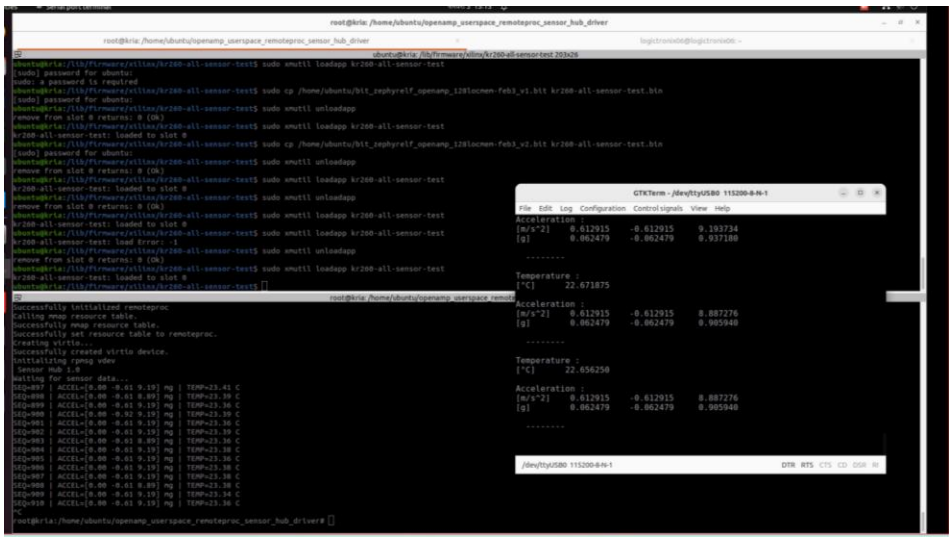
- For downloading and installing kernel, firmware and host code files, please follow this readme - Kria-Ubuntu/README.md

E. Run the Application

Please follow this readme for running the application at KR260 Ubuntu (host app) and KR260 Zephyr (sensor fusion app) - Kria-Ubuntu/README.md

Here are also the steps brief:

1. You can connect the KR260 via SSH (for multi terminal access) as like here:



2. Unload the Kria app and re-load it. This process re-loads the Bitstream as well as Zephyr application (***zephyr.elf***) inside it.
3. Now, run the OpenAMP host app on the Kria-Ubuntu terminal.
4. Here, you must able to see the output as like in above picture.

Revision History

S.N	Revision Number	Revision Details	Author
1.	v1.0	Initial Revision	Krishna Gaihre and Sanam Shakya